Technical Document #660: Software Engineering - Comprehensive Overview

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Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Microservices architecture structures applications as independently deployable services.
- Refactoring improves code structure without changing behavior.
- Code review examines code changes before merging.